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Occupational Safety & Health Training



Healthcare
Personal Protective Equipment



Personal protective equipment (PPE) helps prevent the spread of infectious materials in hospitals, medical clinics, and dental clinics. The use of PPE can protect people and health care workers from infections. This course takes a closer look at the PPE necessary for healthcare workers to protect themselves from many hazards.

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OSHAcademy Course 178 Study Guide

Healthcare: Personal Protective Equipment

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Contact OSHAcademy to arrange for use as a training document.

This study guide is designed to be reviewed off-line as a tool for preparation to successfully complete OSHAcademy Course 178.

We hope you enjoy the course and if you have any questions, feel free to email or call:

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Course Introduction

Personal protective equipment (PPE) is special equipment you wear to minimize exposure to hazards that cause serious workplace injuries and illnesses. Wearing PPE reduces the chance of touching, being exposed to, and spreading infectious materials.

PPE helps prevent the spread of infectious materials in hospitals, medical clinics, and dental clinics. The use of PPE can protect people and health care workers from infections.

All healthcare workers, patients, and visitors should use PPE when there is the potential exposure to blood or other bodily fluids.

This course takes a closer look at the PPE necessary for healthcare workers to protect themselves from many hazards.

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Module 1: Personal Protective Equipment

The use of personal protective equipment (PPE) by healthcare workers helps to reduce exposure to potential hazards in the workplace. Your employer is required perform an assessment to look for potential hazards in the workplace and act to reduce or eliminate them. One of the ways employers can do this is through the use of PPE.

Possible Solutions

There are many ways healthcare workers can effectively use PPE protect themselves from hazards.

Typical employer-provided PPE that employees often use:

- protection for the eyes, face, head, and extremities
- protective clothing
- respiratory devices
- protective shields and barriers

The following are employer responsibilities regarding PPE use:

- PPE (e.g., gloves, goggles, and splash aprons) must be provided as appropriate when exposed to hazards such as cleaning detergents, chemicals, and hazardous drugs. [29 CFR 1910.132(a)]
- Medical Services and First Aid: Where the eyes or body of any person may be exposed to injurious corrosive materials, suitable facilities for quick drenching or flushing of the eyes and body shall be provided within the work area for immediate emergency use. [29 CFR 1910.151(c)]
- Hand Protections: Employers shall select and require employees to use appropriate hand protection when employees' hands are exposed to hazards such as those from skin absorption of harmful substances; severe cuts or lacerations; severe abrasions; punctures; chemical burns; thermal burns; and harmful temperature extremes. [29 CFR 1910.138(a)]
- Employees must be provided with information about the types, proper use, location, removal, handling, decontamination, and disposal of personal protective equipment. [29 CFR 1910.1030(g)(2)(vii)(G)]

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Quiz Instructions

After each section, there is a quiz question. Make sure to read the material in each section to discover the correct answer to these questions. Circle the correct answer. When finished, go online to take the final exam. This exam is open book, so you can use this study guide.

1. Who is responsible to conduct a workplace assessment for potential hazards that may require PPE?

a. Safety committees
b. All employees
c. The employer
d. Third-party consultants

Choose the Right PPE

You may need to use different types of PPE to accommodate different people. Your workplace must have written instructions about when to wear PPE and what type to use. Ask your supervisor how you can learn more about personal protective equipment.

Proper Fit

All personal protective equipment should be safely designed and constructed and should be maintained in a clean and reliable fashion. It should fit comfortably, encouraging worker use. If the personal protective equipment does not fit properly, it can make the difference between being safely covered or dangerously exposed. When engineering, work practice, and administrative controls are not feasible or do not provide sufficient protection, employers must provide PPE to their workers and ensure its proper use. Employers must provide training before employees begin using PPE. This training should include:

- why PPE is necessary
- when PPE is necessary
- what kind of PPE necessary
- how to properly put it on, adjust, wear, and take PPE off

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- PPE limitations
- proper care, maintenance, useful life, and disposal of the PPE

If PPE is to be used, a PPE program should be implemented. This program should address the following:

- hazards present
- the selection, maintenance, and the use of PPE
- training of employees
- monitoring of the program to ensure its ongoing effectiveness

2. Who is required to provide all employees with appropriate PPE?

a. OSHA
b. The safety committee
c. The employer
d. Supervisors

PPE and the Bloodborne Pathogens Standard

PPE is required by the Bloodborne Pathogens Standard if exposure to blood and other potentially infectious materials (OPIM) is anticipated and where occupational exposure remains, after implementing engineering and work practice controls. PPE can provide some protection from infectious materials as a barrier to protect skin and mucus membranes from contact with blood and other potentially infectious materials. [29 CFR 1910.1030(d)(2)(i)]

Appropriate PPE must be provided by the employer, cleaned and laundered and disposed of, at no cost to the employee, in appropriate sizes and used by personnel if blood exposure is anticipated. PPE includes gloves, gowns, laboratory coats, masks, face shields, eye protection, mouthpieces, resuscitation bags, pocket masks, or other ventilation devices. 29 CFR 1910.1030(d)(3)(i)

- Employees must wear gloves when hand contact with blood, mucous membranes, OPIM, or non-intact skin is anticipated, and when performing vascular access procedures, or when handling contaminated items or surfaces [29 CFR 1910.1030(d)(3)(ix)].

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- Employees must wash hands immediately or as soon as feasible after removing gloves or other PPE [29 CFR1910.1030(d)(2)(v)].

- The employer shall ensure that employees who have contact with contaminated laundry wear protective gloves and other appropriate personal protective equipment [29 CFR 1910.1030(d)(4)(iv)(B)].

- PPE will be considered "appropriate" only if it does not permit blood or other potentially infectious materials to pass through to or reach the employee's work clothes, street clothes, undergarments, skin, eyes, mouth, or other mucous membranes under normal conditions of use and for the duration of time which the protective equipment will be used. [29 CFR 1910.1030(d)(3)(i)]

3. According to OSHA, when should healthcare workers wear gloves?

a. During their entire shift
b. When contact with bloodborne pathogens is anticipated
c. OSHA does not require healthcare workers to wear gloves
d. Only when a patient requests them

Types of PPE

Many different types of PPE are used in a healthcare setting. Let's take a closer look.

Gloves

Wearing gloves protects your hands from infectious material and helps reduce the spread of germs. They create a barrier between infectious material and your hands. Healthcare workers should wear gloves every time they anticipate the potential to come in contact with blood, bodily fluids, bodily tissues, mucous membranes, or broken skin. Make sure to wear gloves for this sort of contact, even if a patient seems healthy and has no signs of any germs.

Choosing the Right Type of Gloves

Containers of disposable gloves should be available in any room or area where patient care takes place.

Gloves come in different sizes, so make sure you choose the right size for a good fit.

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- If the gloves are too big, it is hard to hold objects and easier for infectious material to get inside your gloves.
- Gloves that are too small are more likely to rip.

Some cleaning and care procedures require sterile or surgical gloves. Sterile means "free from germs." These gloves come in numbered sizes (5.5 to 9). Know your size ahead of time.

If you will be handling chemicals, check the safety data sheet to see what kind of gloves you will need. Some glove materials do not effectively block chemicals from passing through and contacting the skin.

DO NOT use oil-based hand creams or lotions unless they are approved for use with latex gloves.

If you have a latex allergy, use non-latex gloves and avoid contact with other products that contain latex. (See OSHAcademy course 170 Healthcare: Latex Allergy for more detailed information on latex allergies.)

Removing Gloves

When you take gloves off, make sure the outsides of the gloves do not touch your bare hands. Follow these steps:

1. Using your left hand, grab the outer side of your right glove at the wrist.
2. Pull toward your fingertips. The glove will turn inside out.
3. Hold onto the empty glove with your left hand.
4. Put 2 right-hand fingers in your left glove.
5. Pull toward your fingertips until you have pulled the glove inside out and off your hand. The right glove will be inside the left glove now.
6. Throw the gloves away in an approved waste container.

Always use new gloves for each patient. Wash your hands between patients to avoid passing germs.

Here is a great video on how to properly remove exam gloves.

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4. When taking gloves off, make sure the outsides of the glove do not touch your _____.

a. arm
b. shoulder
c. bare hands
d. elbow

Masks

- Masks cover your mouth and nose.
- Some masks have a see-through plastic part that covers your eyes.
- A surgical mask helps stop germs in your nose and mouth from spreading. It can also keep you from breathing in some germs.
- A special respiratory mask (respirator) forms a tight seal around your nose and mouth. It may be needed so that you do not breathe in small germs like tuberculosis bacteria, measles, chickenpox viruses, or other respiratory viruses.

Eye Protection

Eye protection includes face shields and goggles. These protect the mucous membranes in your eyes from blood and other bodily fluids. If these fluids encounter the eyes, germs in the fluid can enter the body through the mucous membranes.

Face shields and goggles may also limit hand contact with the eyes, providing additional protection from contact exposure to pathogens.

5. A _____ mask forms a tight seal around your nose and mouth, preventing germs from entering your lungs.

a. surgical
b. respiratory
c. plastic
d. nasal

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Protective Clothing

Protective clothing includes gowns, aprons, lab coats, head covering, and shoe covers. This PPE is mean to reduce or eliminate exposure to anticipated hazards that you may encounter in a healthcare setting, such as bloodborne pathogens or chemicals.

- These are often used during surgery to protect you and the patient.
- Visitors wear gowns if they are visiting a person who is in isolation due to an illness that can be easily spread.

You may need special PPE when handling some cancer drugs. This equipment is called cytotoxic PPE.

- You may need to wear a gown with long sleeves and elastic cuffs. This gown should keep liquids from touching your skin.
- You may also need to wear shoe covers, goggles, and special gloves.

Disposal of Protective Clothing

- Employees must remove protective clothing before leaving the room and dispose of it in an appropriately designated area or container for storage, washing, decontamination, or disposal. Employees should never take protective clothing home to be washed or cleaned. [29 CFR1910.1030(d)(3)(vii)] [29 CFR 1910.1030(d)(3)(viii)].

Hearing Protection

Occupational exposure to high noise levels can be found in different areas of the hospital (e.g., laundry, engineering, and heliport). Employee exposure to noisy machinery, or equipment, may induce hearing loss, hearing impairment, hypertension, elevated blood pressure levels, and/or other health hazards.

Ear plugs or muffs should be worn whenever exposure to loud sounds is anticipated. Your employer may also have a hearing protection program to evaluate noise exposure and determine PPE requirements.

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6. What is the purpose of protective clothing PPE?

a. To help identify the type of work employees perform
b. To protect employees from excessive noise
c. To guarantee protection from all hazards
d. To reduce or eliminate exposure to anticipated hazards

Respirators

The hospital environment contains respiratory hazards such as bacteria, viruses, and chemicals that may be inhaled by healthcare workers and patients and cause injury or illness.

A respirator is a personal protective device that is worn on the face, covers at least the nose and mouth, and is used to reduce the wearer's risk of inhaling hazardous airborne particles (including dust particles and infectious agents), gases, or vapors.

Your employer should have a respiratory program that identifies hazards and determines PPE requirements.

Respirator Program Considerations

- Complete annual training and fit testing.
- Know fit tested model and size of respirator.
- Follow manufacturer recommended procedures for donning and doffing.
- Perform the manufacturer recommended seal check method for respirator model.
- Know your respirator program administrator contact.
- Consult Infection Control and Safety Officer.
- Ensure needed PPE is available and located near point of use.
- Know how to identify NIOSH-approved respirator.

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7. What does a respirator help reduce employee exposure to?

a. Excessive noise
b. All forms of radiation
c. Airborne infectious agents
d. Extreme heat or cold

General PPE Guidelines

- Wear gloves, when handling chemicals and/or body fluids.
- Wear safety shoes/boots/covers if hazardous substance is likely to splash.
- Wear an apron/gown/coveralls, if hazardous substance is likely to splash.
- Use a respirator when hazardous substance is airborne such as tuberculosis.
- Wear hearing protection for loud noises such as from equipment.
- Remove PPE carefully to avoid contaminating yourself.
- Dispose of PPE in designated containers before leaving area.

8. What is the primary reason you should be careful when removing PPE?

a. To avoid looking unprofessional
b. To avoid damaging used PPE
c. To avoid contaminating others
d. To avoid contaminating yourself

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References

1. “Healthcare Wide Hazards: Personal Protective Equipment.” Occupational Safety and Health Administration (OSHA). https://www.osha.gov/SLTC/etools/hospital/hazards/ppe/ppe.html

2. “Wearing Gloves in the Hospital.” Medline Plus. https://medlineplus.gov/ency/patientinstructions/000452.htm

3. “Healthcare Wide Hazards: Noise.” Occupational Safety and Health Administration (OSHA). https://www.osha.gov/SLTC/etools/hospital/hazards/noise/noise.html

4. “Hospital Respiratory Protection Program Toolkit.” The National Institute for Occupational Safety and Health (NIOSH). https://www.cdc.gov/niosh/docs/2015-117/pdfs/2015-117.pdf

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